

Superinfections Can Induce Evolutionarily Stable Coexistence of Pathogens

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Within-Host Scenario

- Pathogen strains compete for a single limited resource: the target cells.

The Pessimisation Principle (Diekmann and Planqué [1], Sec. 10.2)

If the population growth rate is a monotone increasing function of a 1-D variable that fully describes the environmental condition, then the evolutionary winner is the trait that **minimises** this variable.

- The optimal **reproduction rate** p minimise the availability of uninfected target cells.

Between-Host Scenario

- Pathogen strains compete for susceptible hosts via transmission, described by a Susceptible-Infected (SI) model.

The SI Model (Diekmann and Planqué [1], Sec. 9.6, Modified)

$$\begin{aligned}\frac{dS}{dt} &= b - \delta S - \beta SI, \\ \frac{dI}{dt} &= -\delta I + \beta SI - \alpha I,\end{aligned}$$

where:

- b : population birth rate;
- α : **virulence**;
- δ (μ in book): death rate related to other causes;
- β : transmission rate (assumed by $\beta = \beta(\alpha)$).

Superinfections

- These two natural selection levels may appear different evolutionarily stable traits (Gilchrist and Coombs [2]).

Superinfections

A new strain can infect an already infected host and (with probability $\phi(p, q)$) immediately take over.

Problem

- 1 How will superinfection change the evolution of the reproduction rate?
- 2 Can we get evolutionarily stable coexistence?

Within-Host Model

Within-Host Dynamics

$$\begin{aligned}\frac{dT}{dt} &= \lambda - kVT - dT, \\ \frac{dT^*}{dt} &= kVT - (\mu(p) + d)T^*, \\ \frac{dV}{dt} &= pT^* - kVT - cV,\end{aligned}$$

where:

- T : uninfected target cells;
- T^* : infected target cells;
- V : free pathogens;
- $\mu(p)$: death rate of T^* .

Within-Host Model

- A stable infected equilibrium $(\hat{T}(p), \hat{T}^*(p), \hat{V}(p))$:

$$\begin{aligned}\hat{T} &= \frac{c}{k(\mathcal{B}_0(p) - 1)}, \\ \hat{T}^* &= \frac{\mathcal{B}_0(p)}{p} \left(\lambda - \frac{cd}{k(\mathcal{B}_0(p) - 1)} \right), \\ \hat{V} &= \frac{\lambda}{c} (\mathcal{B}_0(p) - 1) - \frac{d}{k},\end{aligned}$$

when the within-host basic reproduction ratio

$$\mathcal{R}_0^w(p) = \frac{k\lambda}{k\lambda + dc} \mathcal{B}_0(p) > 1.$$

- Within-host competition leads to lower $\hat{T}(p)$, equivalently maximize $\mathcal{R}_0^w(p)$.

Single Infection Between-Host Model

- Assumption: simple infection model, i.e., cross-immunity.
- Virulence $\alpha(p)$ and transmission $\beta(p)$ mechanistically related to within-host equilibrium:

$$\alpha(p) = A\left(\hat{T}(p), \hat{T}^*(p), \hat{V}(p)\right), \quad \beta(p) = B\left(\hat{T}(p), \hat{T}^*(p), \hat{V}(p)\right).$$

The refined SI Model with trait p and trait q

$$\begin{aligned} \frac{dS}{dt} &= b - \delta S - \beta(p)SI_p - \beta(q)SI_q, \\ \frac{dI_p}{dt} &= -\delta I_p + \beta(p)SI_p - \alpha(p)I_p, \\ \frac{dI_q}{dt} &= -\delta I_q + \beta(q)SI_q - \alpha(q)I_q. \end{aligned}$$

Single Infection Between-Host Model

- Trait q in resident p has **invasion exponent**:

$$s_p(q) = \beta(q)\hat{S}(p) - (\alpha(q) + \delta),$$

with

$$\hat{S}(p) = \frac{\alpha(p) + \delta}{\beta(p)},$$

when

$$\mathcal{R}_0(p) = \frac{b}{\delta} \cdot \frac{\beta(p)}{\alpha(p) + \delta} > 1.$$

- Evolution tends to maximize $\mathcal{R}_0(p)$, equivalently minimize $\hat{S}(p)$.

Superinfection Model

Superinfection Model

$$\begin{aligned}\frac{dS}{dt} &= b - \delta S - \beta(p)SI_p - \beta(q)SI_q, \\ \frac{dI_p}{dt} &= -\delta I_p + \beta(p)SI_p - \alpha(p)I_p + \Phi(q, p)I_p I_q, \\ \frac{dI_q}{dt} &= -\delta I_q + \beta(q)SI_q - \alpha(q)I_q + \Phi(p, q)I_p I_q,\end{aligned}$$

where

$$\Phi(p, q) = \beta(q)\phi(p, q) - \beta(p)\phi(q, p),$$

and the **superinfection function** $\phi(p, q) = \psi\left(\hat{T}(p), \hat{T}(q)\right)$ describes the ability of trait q to take over a host infected by trait p .

Superinfection Model

- Assumption: within-host dynamics is **fast** compared to the dynamics at the host population level.
- **Constraint H_1** : $\psi(x, y) = 0$ whenever $y \geq x$.
A worse (or equal) within-host competitor cannot establish in an already-infected host.
- Invasion exponent:

$$r_p(q) = \beta(q)\hat{S}(p) - (\alpha(q) + \delta) + \Phi(p, q)\hat{l}_p,$$

where

$$\hat{l}(p) = \frac{b}{\beta(p)\hat{S}(p)} - \frac{\delta}{\beta(p)}.$$

Three Cases

Cases	Behaviour of $\psi(x, y)$ when $y \uparrow x$	Consequence
“Jump”	has a jump discontinuity	within-host evolution dominates
“Mechanistic”	continuous but not differentiable	both levels contribute
“Smooth”	differentiable	same as ignoring superinfections

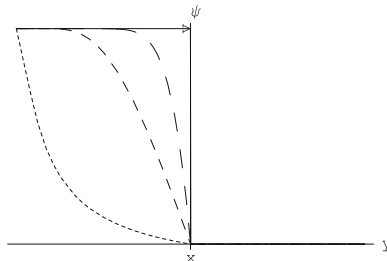


Figure: solid line: “jump” case; two dashed lines: “mechanistic” case; bottom dashed line: “smooth” case.

“Mechanistic” Case

- Initially, the invading trait q is in small quantities, and may go extinct due to demographic stochasticity.
- Modelled by a stochastic birth-death process, the resulting takeover probability $\phi(p, q)$ is continuous but not differentiable at $q = p$.
- When the within-host continuously stable strategy (CSS, a local attractor with respect to the adaptive dynamics) P^* and the between-host CSS p^* are unique, the convergence stable strategy p^{**} lies in between.

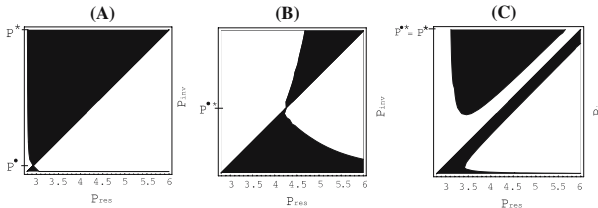


Figure: pairwise invasibility plot (PIP) for (A) “smooth” case; (B) “mechanistic” case; (C) “jump” case. $p^* = p_{max}$, p_{res} = white region, p_{inv} = black region.

Result 1

- Recall: $r_p(q) = \beta(q)\hat{S}(p) - (\alpha(q) + \delta) + \Phi(p, q)\hat{l}_p$,

$$\left. \frac{\partial r_p(q)}{\partial q} \right|_{q=p} = -\beta(p) \left[\left. \frac{d\hat{S}}{dq} \right|_{q=p} - D(\hat{T}(p)) \hat{l}(p) \left. \frac{d\hat{T}}{dq} \right|_{q=p} \right].$$

- When $\hat{S}'(p) = 0$,

$$\text{sign} \left. \frac{\partial r_p(q)}{\partial q} \right|_{q=p} = -\text{sign} \left. \frac{d\hat{T}}{dq} \right|_{q=p},$$

and we move from $p^\bullet$ to $p^{\bullet*}$, in the direction of decreasing $\hat{T}$.

Result 1

Superinfection may lead to an increased virulence (for reasonable $\alpha(p) = A(\hat{T}(p), \hat{T}^*(p), \hat{V}(p))$).

No Optimisation Principle

- The environment is no longer 1-D; we should not expect a single optimal quantity (Metz, Mylius, and Diekmann [3]).

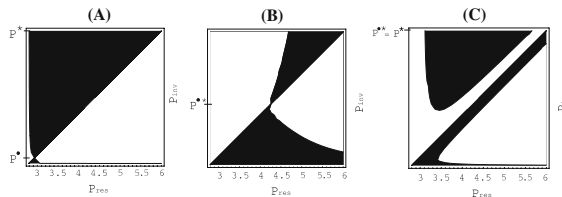


Figure: (B) “mechanistic” and (C) “jump” lack of skew symmetry.

- Such lack of skew symmetry implies **mutual invasibility** $r_p(q) > 0$ and $r_q(p) > 0$ for some pairs (p, q) , yielding **protected dimorphism**.

Branching Point

- Converging dimorphism: near a CSS that is uninvadable, dimorphisms are typically transient.
- In the mechanistic case, the monomorphic attractor $p^{\bullet*}$ can become a (asymmetric) **branching point** (convergence stable but invadable).

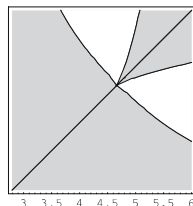
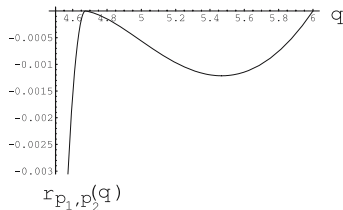


Figure: white areas: mutually invadable; grey areas: opposite signs.

Result 2

Example

When the reinfection dose parameter is large (say $\psi = \psi_{200}$), the monomorphic attractor $p^{\bullet*} \approx 4.69345$ becomes invadable (branching). Simulations suggest an **evolutionarily stable coalition** $(p_1, p_2) \approx (4.675, 6)$: convergence stable and uninvadable.



Result 2

Superinfection can turn $p^{\bullet*}$ into a branching point, leading to evolutionarily stable coexistence of different strains.

Conclusion

- 1 Superinfection may lead to increased virulence.
- 2 Superinfection may lead to evolutionarily stable coexistence of different strains.

Discussion

Potential Usage

- Chronic infections that require reinfection: HIV, Hepatitis B, and Hepatitis C virus.
- Evolutionary dynamics framework for: SEI, SIR models, or different incidence rates.

Limitations & Future Work

- Rare mutations and SI model with no immune clearance.
- Within-host dynamics are faster, and the new strain takes over immediately.
- Simple infection model in the between-host model.

Reference

- [1] Odo Diekmann and Bob Planqué. **Mathematical Methods to Gain Biological Insight**. Lecture notes for the MasterMath course Mathematical Biology. 2019. URL: <https://www.few.vu.nl/~rplanque/resources/syllabus.pdf>.
- [2] Michael A. Gilchrist and Daniel Coombs. “Evolution of Virulence: Interdependence, Constraints, and Selection Using Nested Models”. In: **Theoretical Population Biology** 69.2 (2006), pp. 145–153. DOI: 10.1016/j.tpb.2005.07.002.
- [3] Johan A. J. Metz, Sido D. Mylius, and Odo Diekmann. **When Does Evolution Optimize? On the Relation Between Types of Density Dependence and Evolutionarily Stable Life History Parameters**. IIASA Working Paper WP-96-004. Laxenburg, Austria: International Institute for Applied Systems Analysis, 1996. URL: <https://pure.iiasa.ac.at/5021/>.
- [4] Barbara Boldin and Odo Diekmann. “Superinfections Can Induce Evolutionarily Stable Coexistence of Pathogens”. In: **Journal of Mathematical Biology** 56.5 (2008), pp. 635–672. DOI: 10.1007/s00285-007-0135-1.